

Matrix Ring :-

Let R be any ring and $n \geq 1$ be an integer.

Let $S = M_n(R) = \{ (a_{ij}) \mid a_{ij} \in R, 1 \leq i, j \leq n \}$,

Set of $n \times n$ matrices with entries in R .

$$a_{ij} = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

$$(a_{ij}), (b_{ij}) \in M_n(R), (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$$

$$(a_{ij})(b_{ij}) = c_{ij}, \text{ where } c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \forall i, j, 1 \leq i, j \leq n$$

$$\left\{ \begin{pmatrix} c_{11} & \dots & c_{1n} \\ c_{21} & \dots & c_{2n} \\ \vdots & \ddots & \vdots \\ c_{n1} & \dots & c_{nn} \end{pmatrix} \right\}$$



If $1 \in R$, $1 = 1_S = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = I_n$

If $a_{ij} = 0$, $\forall i \neq j$, then we call (a_{ij}) as a diagonal matrix, denoted by $\text{diag}(a_1, a_2, \dots, a_n)$

If $a_{11} = \dots = a_{nn} = a$ (say), then $\text{diag}(a, \dots, a)$ is called a scalar matrix.

$\text{diag}(a, \dots, a) = a I_n$ if $1 \in R$

$$\begin{pmatrix} a & 0 & \dots & 0 \\ 0 & a & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a \end{pmatrix} = a \underbrace{\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}}_{I_n}$$

Remark.

If R is non-Commutative, then S is non-Commutative.

Pf. Suppose R is non-Commutative.

$\exists a, b \in R$ s.t. $ab \neq ba$

$\text{diag}(a, \dots, a) \text{diag}(b, \dots, b)$

$$= \begin{pmatrix} a & 0 & \dots & 0 \\ 0 & a & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a \end{pmatrix} \begin{pmatrix} b & 0 & \dots & 0 \\ 0 & b & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & b \end{pmatrix}$$

$$= \begin{pmatrix} ab & 0 & \dots & 0 \\ 0 & ab & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & ab \end{pmatrix}$$

$= \text{diag}(ab, \dots, ab) \neq \text{diag}(ba, \dots, ba)$

$= \text{diag}(b, \dots, b) \text{diag}(a, \dots, a)$

If $1 \in R$, then $a I_n \cdot b I_n = ab I_n$
 $\neq ba I_n$
 $= b I_n \cdot a I_n$

Notation

For $a \in R$, $a E_{ij} = \begin{pmatrix} 0 & \dots & 0 \\ & a & \\ 0 & \dots & 0 \end{pmatrix}$ $\leftarrow$ i th row

$E_{11} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ & 0 & & \\ & & \ddots & \\ 0 & & & 0 \end{pmatrix}$

$a E_{ij} b E_{kl} = \begin{cases} ab E_{il} & , \text{ if } j=k \\ 0 & , \text{ otherwise, } \forall a, b \in R \end{cases}$

Remark -

Even if R is commutative, S need not be commutative

(unless R is a trivial ring) $[xy = 0, \forall x, y \in R]$.

Pf - Take $a, b \in R$ s.t. $ab \neq 0$

Take $A = a E_{12}$, $B = b E_{11}$

$AB = a E_{12} b E_{11} = 0$

$BA = b E_{11} a E_{12} = ba E_{12}$

$= ab E_{12}$

$\neq 0$

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Theorem.

Let I be an ideal of a ring R . Then I is a 2-sided ideal in $R \iff I$ is the kernel of some homo. $f: R \rightarrow S$ for a suitable ring S .

Proof - Suppose I is a 2-sided Ideal of R . $I \leq R$.

Take $S = R/I$ [Already $(R/I, +, \cdot)$ is a ring. $(a+I) + (b+I) = (a+b)+I$
 $(a+I) \cdot (b+I) = (ab)+I$]

Define $f: R \rightarrow R/I$ by $f(x) = x+I$.

f is homo:

$$f(x+y) = \dots = f(x) + f(y)$$

$$f(xy) = \dots = f(x)f(y)$$

$$\ker f = \{x \in R \mid f(x) = 0\}$$

$$= \{x \in R \mid x+I = 0 \text{ in } R/I\}$$

$$= \{x \in R \mid x-0 \in I\}$$

$$= \{x \in R \mid x \in I\}$$

$$= I$$

Conversely, suppose that R and S are rings.

$f: R \rightarrow S$ homomorphism.

let $I = \ker f$

To show: I is a 2-sided ideal of R .

Take $x, y \in I$

$$f(x) = 0 = f(y)$$

$$f(x-y) = f(x) - f(y)$$

$$= 0$$

$$\Rightarrow x-y \in \ker f$$

Take $\pi \in R, x \in I$

$$\begin{aligned} f(\pi x) &= f(\pi)f(x) \\ &= f(\pi) \cdot 0 \\ &= 0 \end{aligned}$$

$$\Rightarrow \pi x \in \ker f$$

Similarly $x\pi \in \ker f$

So $\ker f = I$ is an ideal of K

Remark.

Suppose $f: K \rightarrow S$ and $g: S \rightarrow T$ are homomorphisms.

Then $\ker(g \circ f) \supseteq \ker f$

$$\{x \in K \mid f(x) = 0\} \subseteq \{x \in K \mid g(f(x)) = 0\}$$

$$\{x \in K \mid f(x) = 0\} \subseteq \{x \in K \mid g(f(x)) = 0\}$$

$$I \subseteq \ker(g \circ f)$$

Conversely, suppose that R and S are rings.

$f: R \rightarrow S$ homomorphism.

$$\ker f = I$$

To show: I is a 2-sided ideal of R .

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Theorem -

Let G be a finite group; S, T are subgroups of G .

Then $|ST| = \frac{|S| |T|}{|S \cap T|}$

$$ST = \{st \mid s \in S, t \in T\}$$

Proof - Consider $S \times T = \{(s, t) \mid s \in S, t \in T\}$

Define a relation $\sim$ on $S \times T$ by

$$(s, t) \sim (s', t') \iff \exists a \in S \cap T \text{ such that } s' = sa, t' = at^{-1}$$

To show: ' $\sim$ ' is an equivalence relation on $S \times T$.

Reflexive -

$$1 \in S \cap T$$

$$s \sim s \cdot 1, t \sim (1)^{-1} t$$

$$\Rightarrow (s, t) \sim (s, t)$$

Symmetry -

Suppose $(s, t) \sim (s', t')$

$$\Rightarrow \exists a \in S \cap T \text{ such that } s' = sa, t' = at^{-1}$$

Now $s = s' a^{-1}$ and $t = (a^{-1})^{-1} t'$, here $a^{-1} \in S \cap T$

$$\therefore (s', t') \sim (s, t)$$

Transitive -

$$(s, t) \sim (s', t') \text{ and } (s', t') \sim (s'', t'')$$

$$\Rightarrow \exists a \in S \cap T \text{ s.t. } s' = sa, t' = at^{-1}$$

$$\exists b \in S \cap T \text{ s.t. } s'' = s'b, t'' = b^{-1} t'$$

Now $s'' = s'b$

$$= s(ab), \text{ here } ab \in S \cap T, \text{ and } s' = sa$$

$$t'' = b^{-1}t'$$

$$= b^{-1}a^{-1}t$$

$$= (ab)^{-1}t$$

$$\therefore (s, t) \sim (s'', t'')$$

Let $\overline{(s, t)}$ or $[(s, t)]$ denote the equivalence class of (s, t) in $S \times T$.

Let $S \times T / \sim$ be the set of all equivalence classes.

$$\overline{(s, t)} = [(s, t)] = \{ (s', t') \mid (s, t) \sim (s', t') \}$$

$$= \{ (s', t') \mid s' = sa, t' = a^{-1}t \}$$

$$= \{ (sa, a^{-1}t) \mid a \in SNT \}$$

$$|\overline{(s, t)}| = |SNT| \quad (\text{each})$$

Also let $S \times T = \bigcup_{i=1}^k \overline{(s_i, t_i)}$ be the disjoint union of equivalence classes.

$$|S \times T| = \left| \bigcup_{i=1}^k \overline{(s_i, t_i)} \right| = |\overline{(s_1, t_1)}| + |\overline{(s_2, t_2)}| + \dots + |\overline{(s_k, t_k)}|$$

$$= |SNT| + |SNT| + \dots + |SNT|$$

$$= k |SNT| \quad \text{①}$$

Define $f: S \times T / \sim \rightarrow SNT$ by

$$f(\overline{(s_i, t_i)}) = s_i t_i$$

f is well defined and injective

$$\overline{(s_i, t_i)} = \overline{(s_j, t_j)} \iff s_j = s_i a, t_j = a^{-1} t_i$$

$$\begin{aligned}
 s_j t_j &= s_i a \cdot a^{-1} t_i \\
 &= s_i e t_i \\
 &= s_i t_i
 \end{aligned}$$

This means $f(\overline{s_i, t_i}) = f(\overline{s_j, t_j})$

Clearly f is surjective.

$$\therefore |S \times T| = |ST|$$

~~From ①~~ $\Rightarrow k = |ST|$ (Since there are k equivalence classes)

$$\{(1, 2) \sim (1, 2) \mid (1, 2)\} = [(1, 2)] = \overline{(1, 2)}$$

From ①,

$$\{1' \cdot 2 = 1, 2 = 2 \mid (1, 2)\} =$$

$$|S \times T| = |ST| |SNT|$$

$$\{T \cap 2 \neq \emptyset \mid (1' \cdot 2, 2)\} =$$

$$\Rightarrow |ST| = \frac{|S \times T|}{|SNT|} = \frac{|S| |T|}{|SNT|}$$

Alternatively

Def-

A commutative integral domain R (with or without unity) is called a Euclidean domain, if there is a map -

Algorithm map $\boxed{d: R^* \rightarrow \mathbb{Z}^+, \text{ where } R^* = R \setminus \{0\} \text{ such that -}}$

① $\forall a, b \in R^*, a|b \Rightarrow d(a) \leq d(b)$

equivalently $d(x) \leq d(xy)$

② $\forall a \in R, b \in R^*, \exists q, r \in R$ (depending on a & b)

such that $a = qb + r$, either $r = 0$, $d(r) < d(b)$

division algorithm

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Ex.

1. Let K be any field. $K^* = K \setminus \{0\}$

Define the algorithm map as a constant map $d: K^* \rightarrow \mathbb{Z}^+$,

$$d(x) = 1, \forall x \in K^*$$

$$(i) d(x) \leq d(xy), \forall x, y \in K^* \quad [\text{here } y=1]$$

$$(ii) \text{ For each } x, x = (xa^{-1})a + 0, \forall a \in K^*, \forall x \in K$$

$$\downarrow \quad \downarrow \quad \swarrow$$
$$b = qa + r$$

2. The ring of integers is an Euclidean domain.

Define algorithm map $d: \mathbb{Z}^* \rightarrow \mathbb{Z}^+$ by

$$d(n) = |n|, \forall n \in \mathbb{Z}^*$$

$$(i) d(x) : |x| \leq |xy| = d(xy), \forall y \in \mathbb{Z}^*$$

$$(ii) \text{ Given } n \in \mathbb{Z} \text{ \& } a \in \mathbb{Z}^*$$

$$|n| = q|a| + r \text{ with } 0 \leq r < |a|, \text{ for unique } q, r \in \mathbb{Z}$$

$$\downarrow \quad \downarrow$$
$$q = qa + r$$

$$\text{i.e. } n = q'a + r', \text{ where } q' = \pm q, r' = \pm r \text{ depend upon } n \text{ \& } a.$$

3. The ring $\mathbb{Z}[i]$ of Gaussian integers is an Euclidean ring.

$$\mathbb{Z}[i] = \{a+ib \mid a, b \in \mathbb{Z}\}$$

Define an algorithm map $d: \mathbb{Z}^*[i] \rightarrow \mathbb{Z}^+$.

$$d(m+in) = |m+in|^2$$

$$= (m+in)(\overline{m+in})$$

$$= (m+in)(m-in)$$

$$= m^2 + n^2, \quad m, n \in \mathbb{Z}$$

$$(i) \quad d(m_1+in_1) \leq d((m_1+in_1)(m_2+in_2))$$

$$m_1^2 + n_1^2 \leq d(m_1 m_2 + i(m_1 n_2 + n_1 m_2) - n_1^2 n_2^2)$$

$$\Rightarrow m_1^2 + n_1^2 \leq d((m_1 m_2 - n_1^2 n_2^2) + i(m_1 n_2 + n_1 m_2))$$

$$(ii) \quad \text{Take } \alpha \in \mathbb{Z}[i] \text{ and } a \in \mathbb{Z}[i]^*$$

To find $q, \pi \in \mathbb{Z}[i]$ such that

$$\alpha = qa + \pi, \text{ either } \pi = 0 \text{ or } |\pi|^2 < |a|^2$$

Consider the complex number, $z = \frac{\alpha}{a}$

Theorem. (The Eisenstein Criterion for irreducibility over $\mathbb{Q}$)

Let $f(x) = a_0 + a_1x + \dots + a_nx^n$ be a polynomial with

integer coefficients. Suppose that for some prime number p divides a_i ($0 \leq i \leq n-1$), p does not divide a_n , $p^2 \nmid a_0$,

then $f(x)$ is irreducible poly. over $\mathbb{Q}$.

Ex-

If p is a prime number, prove that the poly.

$f(x) = x^n - p$, is irreducible over the rationals.

Solⁿ - $f(x) = x^n - p$

$$= a_0 + a_1x + \dots + a_nx^n, \quad a_i = 0 \quad (1 \leq i \leq n-1)$$

$$p \mid a_i, \quad 0 \leq i \leq n-1 \quad \begin{matrix} a_n = 1 \\ a_0 = p \end{matrix}$$

$$p \mid a_0, \quad p \mid a_1 = 0, \quad p \mid a_2 = 0, \quad \dots, \quad p \mid a_{n-1} = 0$$

$p \nmid a_n = 1$, $p^2 \nmid a_0 = p$, All the condition of Eisenstein Criteria satisfies.

$\therefore f(x)$ is irreducible over $\mathbb{Q}$.

By: $f(x) = 16x^5 - 9x^4 + 3x^2 + 6x - 21$ is irreducible over $\mathbb{Q}$ (with $p=3$)

Ex- Prove that the polynomial -

$f(x) = 1 + x + \dots + x^{(p-1)}$ where p is prime, is irreducible over the field of rationals.

Proof - Consider $x \mapsto 1+x$

$$g(x) = f(x+1)$$

$$= 1 + (1+x) + \dots + (1+x)^{p-1}$$

$$= 1 + (x+1) + (x^2+2x+1) + \dots + \left\{ x^{p-1} + \binom{p-1}{1}x^{p-2} + \dots + \binom{p-1}{p-1}x^{(p-1)-(p-1)} \right\}$$

$$= p + [1+2+\dots+(p-1)]x + [1+3+\dots+(p-1)]x^2 + \dots + x^{p-1}$$

$$= p + \left[\frac{p(p-1)}{2} \right] x + \dots + x^{p-1}$$

$$= a_0 + a_1 x + \dots + a_{p-1} x^{p-1}$$

$$\text{where } a_0 = p, a_1 = \frac{p(p-1)}{2}, \dots, a_{p-1} = 1$$

$$\text{Clearly } p \nmid a_0, p \nmid a_1, \dots, p \nmid a_{p-1}, p^2 \nmid a_0$$

By Eisenstein Criterion $g(x)$ is irreducible over $\mathbb{Q}$.

$$x \mapsto x-1$$

$\therefore f(x)$ is irreducible over $\mathbb{Q}$.

Ex- Show that

$$f(x) = x^2 - q \text{ is irreducible over } \mathbb{Z}_p$$

Proof

Suppose $f(x)$ is reducible over $\mathbb{Z}_p$, then $p \mid q$ where p is prime, is irreducible over

Ex- Prove that the poly.

$f(x) = x^2 + x + 1$, is irreducible over the integers modulo 2,

$$\text{i.e. } \mathbb{Z}_2 = \{0, 1\}.$$

Solⁿ- If $f(x)$ is not irreducible, then $x^2 + x + 1 = a(x)b(x)$,

for some poly. $a(x), b(x)$ of deg. 1 over $\mathbb{Z}_2$.

deg 1. poly. over $\mathbb{Z}_2$ are x or $x+1$.

Case-1-

Suppose $a(x) = x$

$$x \cdot b(x) = x(x+1) = x^2 + x$$

$\Rightarrow x$ divides $x^2 + x + 1$, a contradiction.